

MIDTERM (SHORT NOTES)

2024 - 2025

CS201 MCQ's

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Lecture # 01:

What is a program in the context of programming?

- ✓ A sequence of steps to solve a problem
- A random collection of commands
- A set of files and folders
- An error checking method

According to Alan Perlis, why is teaching programming against the grain of modern education?

- ✓ It involves planning, organizing, and being self-critical
- Because it is boring and repetitive
- Because it's expensive
- Because students don't need computers

What does Steve Summit mean by saying computers need detailed instructions?

- ✓ Computers need precise and complete instructions
- Computers are smart and understand human language
- Computers can infer what we mean
- Computers can fix our mistakes

Why is programming considered important?

- ✓ It improves problem-solving and analytical thinking
- It is only for software engineers
- It makes computer games
- It avoids using application software

Which of the following is *not* a skill required for programming?

- Using a mouse

- ✓ Avoiding comments in code
- Paying attention to detail
- Thinking about reusability

Why should code be commented liberally?

- ✓ To explain functionality for yourself and others
- To make the program run faster
- To confuse the compiler
- To hide logic from users

What does it mean when we say “computers are stupid”?

- ✓ They follow instructions exactly without understanding
- They can’t perform arithmetic
- They get viruses easily
- They cannot be programmed

What is the purpose of a program design recipe?

- ✓ To follow structured steps for writing code
- To learn different programming languages
- To find bugs in operating systems
- To build hardware systems

Which step is *not* part of the design recipe mentioned in the lecture?

- ✓ Guessing the output
- Analyzing a problem
- Using examples
- Formulating precise comments

What should be the first step in writing a program?

- ✓ Understand and analyze the problem
- Copy code from the internet
- Start coding immediately
- Create graphics first

Which statement about reusability is true?

- ✓ Reusable code saves time and effort
- Reusable code should always be rewritten
- Reusability is not important
- Comments make code reusable

What should be assumed about users when designing a program?

- They are computer experts
- They can write code
- ✓ They may not understand technical terms
- They can debug errors

Why is it bad to be vague when writing code?

- ✓ The computer cannot infer or guess intentions
- It confuses other programmers

It slows the system
It makes the UI look ugly

What is meant by logical correctness in programming?

- ✓ Code should solve the problem as intended
- Code must run fast
- Code must be grammatically correct
- Code should be complicated

What makes programming fun, according to the lecture?

- ✓ It's a creative way to solve problems
 - It's a way to get rich
 - It's easier than using apps
 - It avoids using logic
-

Lecture # 02:

What are the two main categories of software?

- ✓ System software and Application software
- Hardware and Software
- Compiler and Interpreter
- Operating system and Antivirus

Which of the following is *not* part of system software?

- Operating System
- ✓ Microsoft Word
- Device Drivers
- Utilities

What is the primary function of an operating system?

- To entertain users
- To create games
- ✓ To manage hardware resources
- To browse the internet

What is a device driver?

- A hardware component
- ✓ A program that helps communicate with hardware
- A gaming software
- A virus checker

Why do we need drivers for devices like printers and scanners?

- ✓ To allow the computer to interact with them
- To make them look good
- To prevent hacking
- To store passwords

What does the Disk Defragmentation utility do?

Deletes all files

✓ Rearranges file fragments to speed up access

Compresses files permanently

Installs antivirus software

What is TWAIN associated with?

Monitors

✓ Scanners

Modems

CD-ROMs

What is the main use of application software?

✓ To perform tasks for users

To run hardware

To protect the operating system

To load the BIOS

Which of these is an example of application software?

Device driver

Disk utility

✓ Inventory management system

Operating system

Where was the C language developed?

✓ Bell Laboratories

Microsoft

Apple

MIT

Which languages influenced the creation of C?

✓ BCPL and B

Python and Java

C++ and Java

Assembler and COBOL

What does ANSI stand for in C standardization?

✓ American National Standards Institute

Advanced National Software Institute

Android Networking Standard Interface

Allied Networking Standards International

What is the purpose of a compiler?

To run games

✓ To convert source code into machine code

To hack software

To write documents

Which tool is used to test a program line by line?

Linker

✓ Debugger
Loader
Preprocessor

Why do we need a linker?

To start a program
✓ To combine all used routines and functions
To delete errors
To check grammar

Lecture 3

What is the first function executed in a C program?

✓ `main()`
`start()`
`mainblock()`
`header()`

What does `#include <iostream.h>` do?

✓ Tells compiler to include input/output functionalities
Starts main function
Includes variable list
Ends the program

What does `cout` stand for in C++?

Count operator
✓ Console output stream
Counter utility
Column out function

Which symbol is used to terminate a statement in C?

,
:
✓ ;
.

What happens if you forget a semicolon at the end of a statement?

✓ Compilation error
Logical error
Runtime crash
Nothing happens

Which type of data does `int` store?

✓ Whole numbers
Decimal numbers
Characters
Strings

What is the output of `cout << x; if x = 5; ?`

Memory location of x

Character x

☒ 5

Nothing

What does the assignment operator = do?

Compares values

☒ Stores value to a variable

Checks data type

Runs the program

Which of these is a valid declaration?

`int; x`

`int = x;`

☒ `int x;`

`int x()`

Which data type is used for small integers?

☒ `short`

`tiny`

`char`

`smallint`

What is the size of `int` usually in bytes?

2

☒ 4

6

8

What is the purpose of `float` data type?

To store characters

☒ To store decimal numbers

To store long integers

To hold arrays

Which keyword is used for double precision real numbers?

☒ `double`

`longfloat`

`real`

`precision`

Which symbol is used for modulus in C?

`&`

☒ `%`

`^`

Which of the following gives remainder when dividing 9 by 2?

☒ `9 % 2`

`9 / 2`

```
9 * 2
9 + 2
```

Lecture 4

How many variables were used in the sample average age program?

- ☒ 12
- 10
- 15
- 5

What is the purpose of `cin` in C++?

- ☒ To take input from the user
- To print output
- To assign a value
- To comment code

What happens when `cin >> x;` is executed?

- It prints x
- ☒ It takes input and stores in x
- It deletes x
- It shows error

What does this line do: `AverageAge = TotalAge / 10;`?

- Adds 10 to total
- ☒ Calculates average
- Subtracts 10 from total
- Multiplies 10 with average

Why is the average incorrect in the sample program?

- Too many variables
- Wrong input
- ☒ Integer division truncates decimal part
- Compiler error

Which operator has higher precedence?

- +
- ☒ *

==

How can you force the order of operations in an expression?

- ☒ Using parentheses `()`
- Using semicolons
- Using `++`
- Using spaces

Which is a correct expression for $a(x + b(y + cz^2))$ in C?

`a * (x + b + y + c * z)`

☒ `a * (x + b * (y + c * z * z))`

`a + x + b + y + c + z^2`

`a(x + b(y + cz2))`

Which symbol is used for grouping statements in an expression?

`}`

`[]`

☒ `()`

`||`

Which of the following causes logical error in expression evaluation?

Correct operator usage

☒ Incorrect parentheses placement

Missing header file

Using constants

What is the result of $(10 + 10) * 5$?

10

☒ 100

20

50

How is the remainder of a number obtained in C?

By dividing it

☒ Using `%` operator

Using `&`

Using `+`

What happens if parentheses are placed incorrectly in a complex expression?

☒ Gives wrong result

Gives syntax error

Program won't compile

Nothing changes

How do we reuse a variable in calculations?

Using `&&`

☒ Assign new value to the same variable

Delete old value

Using `@`

What happens when a float is used with `cout`?

☒ Shows decimal value

Only integer part shows

Compiler error

Shows 0

Lecture 5

What is the purpose of conditional statements in programming?

- ✓ To make decisions in code
- To end the program
- To declare variables
- To create loops

Which keyword is used to write a condition in C?

- ✓ if
- when
- do
- loop

What does an if block do if the condition is false?

- ✓ Skips the statements
- Runs all statements
- Shows error
- Repeats the condition

Which operator is used for equality check in C?

- =
- ✓ ==
- equals
- !=

What does the > symbol mean in a condition?

- Less than
- ✓ Greater than
- Equal to
- Multiply

What is a block of statements in C enclosed in?

- ✓ { }
- ()
- []
- ""

Which of these causes a logical error in an if condition?

- Using ==
- Using !=
- ✓ Using = instead of ==
- Using { }

What are relational operators used for?

- Math operations
- ✓ To compare values
- To take input
- To print text

What does `!=` mean in C?

- Equal
- ☒ Not equal
- Greater
- Less

What happens if you forget braces `{}` in an `if` statement with multiple lines?

- ☒ Only the first line is executed
- Compiler error
- All lines run
- None runs

What is the flowchart symbol for a decision?

- ☒ Diamond
- Circle
- Square
- Arrow

Which tool helps visualize decision making in programs?

- Debugger
- ☒ Flowchart
- Compiler
- Loader

What is the result of `if (5 > 2)`?

- False
- ☒ True
- Error
- 0

Which example uses an incorrect conditional syntax?

- `if (x > 5) {}`
- ☒ `if x > 5 {}`
- `if (x == y) {}`
- `if (a <= b) {}`

What is indentation used for in C programming?

- ☒ To improve readability
- To make code faster
- To compile code
- To comment code

Lecture 6

What is the main use of `if/else` in programming?

- ☒ To handle both true and false conditions
- To loop statements

To assign variables
To define functions

How many conditions can an `if` statement check?

One
☒ One at a time
Two
Unlimited

What happens when the condition in `if` is false but `else` is present?

☒ `else` part is executed
Compiler crashes
Loop starts
Nothing happens

What is the output of `if (4 == 4) cout << "Yes"; else cout << "No";`?

Yes
☒ Yes
No
Error

What is the purpose of `else if` in C?

To comment code
☒ To add multiple conditions
To end loop
To assign value

Which of the following shows proper syntax for `if/else`?

`if (x > y) then {}`
☒ `if (x > y) {}`
`if x > y: {}`
`if (x > y) => {}`

What is the logical operator `&&` used for?

OR
☒ AND
NOT
XOR

What does the `||` operator do?

☒ Logical OR
Logical AND
Equals
NOT

What is the result of `if (true || false)`?

False
☒ True
Error
None

What is ! used for in C?

- ✓ NOT
- OR
- AND
- Compare

Which expression returns true?

- ✓ `(x > 0 && y > 0)`
- `(x < 0 && y > 0)`
- `(x != x)`
- `(false && true)`

What does `(a >= 5 || b < 10)` mean?

- a is less than 5
- ✓ Either a is ≥ 5 OR b is < 10
- Both a and b are less than 5
- a must be equal to b

Which is a valid logical expression?

- `(x > 5 AND y < 10)`
- `(x < 5 || y > 2)`
- ✓ `(x < 5 && y > 2)`
- `x == 5 -> y == 3`

What will `!true` return?

- ✓ False
- True
- Error
- 0

What is the outcome of `if (!(5 > 2))`?

- True
- ✓ False
- 5
- 2

Lecture 7

What is a loop used for in programming?

- ✓ Repeating a set of instructions
- Skipping code
- Making decisions
- Declaring variables

Which loop is best when the number of iterations is known?

- while
- ✓ for

if
do

What is the syntax for a for loop?

for x = 0 to 10 {}
for (x = 0; x < 10; x++)
☒ for (int i = 0; i < 10; i++)
loop (i = 0, 10)

Which statement is executed first in a for loop?

Condition
☒ Initialization
Update
Break

What does the condition part of a for loop do?

☒ Checks whether to continue or stop
Starts the loop
Prints output
Resets variable

What is the final part of a for loop?

☒ Update
Initialization
Condition
End

How many times does for (int i = 0; i < 3; i++) run?

0
☒ 3
2
4

What is the use of increment i++ in loops?

☒ Updates the loop control variable
Ends the loop
Skips iteration
Adds to all values

What will i++ do if i = 5?

i = 3
i = 6
☒ Increments i after use
Decrements i

Which loop uses a counter variable usually?

while
☒ for
if
else

What happens if loop condition is always true?

- ☒ Infinite loop
- Stops automatically
- Syntax error
- Skips loop

What happens if initialization is skipped in a loop?

- ☒ Previous value is used
- Loop never runs
- Compiler error
- Nothing runs

What's the result of this loop: `for (int i=0; i<1; i++) cout << i;`

- 0 1
- ☒ 0
- 1
- Nothing

Which of the following is invalid loop syntax?

- `for (int i=0; i<10; i++)`
- ☒ `for int i = 0 to 10`
- `for (i=0; i<10; ++i)`
- `for (int i=5; i<15; i+=2)`

What happens after the update step in a `for` loop?

- ☒ Condition is re-checked
- Program ends
- Variable resets
- Nothing

Lecture 8

Which loop checks condition before executing the body?

- ☒ `while`
- `do-while`
- `for`
- `repeat`

What is the syntax of a `while` loop?

- `while x > 5 {}`
- ☒ `while (x > 5) {}`
- `loop x > 5 {}`
- `when (x > 5)`

When does a `while` loop stop?

- After 5 iterations
- ☒ When condition becomes false
- Never
- On break

What is the condition type used in `while` loop?

- ✓ Boolean expression
- Integer only
- Float only
- No condition

What will happen if the condition is false at the start?

- ✓ Loop never executes
- Executes once
- Runs infinitely
- Skips to else

What is the difference between `while` and `do-while`?

- `while` always runs
- ✓ `do-while` runs at least once
- `do-while` is faster
- No difference

Which of these loops ensures one-time execution even if the condition is false?

- `while`
- ✓ `do-while`
- `for`
- `if`

Syntax of `do-while` loop ends with?

- `if`
- ✓ `while (condition) ;`
- `then`
- `loop`

What happens if there is a `;` after `while (condition) ;`?

- ✓ It completes the loop
- Compiler error
- Syntax error
- Infinite loop

When is `while` preferred over `for`?

- ✓ When repetitions are unknown
- When values are static
- When no update is needed
- When conditions are fixed

What does the loop body contain?

- Variable declaration
- ✓ Repeated code
- Header
- Only print statements

What can break a loop?

- `continue`

✓ break
new
skip

What does `continue` do in a loop?

Ends loop
✓ Skips current iteration
Repeats loop
Stops all code

What happens if the loop condition is always true?

✓ Infinite loop
Ends at once
Compiler error
Prints 0

What is the output of `int i = 10; while(i < 5) cout << i;?`

✓ Nothing
10
5
0

Lecture 9

What is the main purpose of using functions in programming?

✓ To break code into smaller, reusable modules
To speed up compilation
To allocate memory
To store variables

Which of these is a function declaration in C?

✓ `int square(int);`
`int square { ... };`
`square(int x);`
`function square(int);`

Which part of a function includes its return type, name, and parameter list?

✓ Function signature (declaration/prototype)
Function body
Main function only
Compiler directive

What term describes the code block inside braces following a function header?

Function prototype
✓ Function definition (body)
Header file
Macro

Which statement correctly calls a function named `square` with argument `x` and stores its result?

`square(x);`
`call square(x);`
☒ `result = square(x);`
`int result; square(x);`

What happens if you call a function before declaring its prototype?

☒ Compiler error or implicit declaration
Function runs twice
Automatically inlined
Variable overwritten

Which of these statements about `main()` is true?

It cannot call other functions
☒ It is the program's entry point
It must not return a value
It always takes no arguments

Which part of a function can be omitted if the full definition appears before use?

☒ Separate function prototype
Braces around the body
Return statement
Parameter names

Which category of functions does not return any value?

Inline functions
Recursive functions
☒ Void functions
Lambda functions

When is the body of a function executed?

On program start automatically
☒ Only when the function is called
When the file is compiled
At program end

What design approach divides a task like “making a stool” into subtasks?

Bottom-up design
Inside-out design
Sideways design
☒ Top-down design

What would the prototype be for a function that takes a `double` and returns an `int`?

`double func(int);`
`int func();`
☒ `int func(double);`
`void func(double);`

Which of the following is an advantage of using functions?

- Longer code length
- Reduced modularity
- Harder debugging
- ✓ Better code reuse and readability

Which is a correct function header in C?

- `square(int x) { ... }`
- ✓ `int square(int x) { ... }`
- `int square x { ... }`
- `square<int x> { ... }`

How do you return the square of n from within a function?

- Subtraction only
- ✓ `return n * n;`
- `print n * n;`
- `exit(n * n);`

Lecture 10

What does the `#include` directive do?

- Begins a loop
- Defines a variable
- Ends a file
- ✓ Inserts contents of a header file into the program

Which directive is commonly used to define symbolic constants?

- `const`
- `static`
- `extern`
- ✓ `#define`

Which scope allows a variable to be accessed from any function/file?

- Function scope
- Block scope
- Class scope
- ✓ Global scope

Which scope limits a variable to a single function?

- ✓ Function-level scope
- Global scope
- Block-level scope
- File-level scope

What does block-level scope refer to?

- ✓ Variables declared inside `{ }`
- Functions only
- Global definitions
- Parameter lists

Why minimize the use of global variables?

- They are faster
- To make code longer
- Because compilers disallow them
- ✓ To reduce unintended side effects

In C's default calling mechanism, arguments are passed how?

- By reference
- By pointer
- By global variable
- ✓ By value

What is a consequence of call by value?

- Functions cannot return values
- Memory leaks
- ✓ Original variables remain unchanged
- Variables can be modified inside the caller

Which mechanism allows a function to modify the caller's variable?

- Loops only
- ✓ Call by reference
- Call by value
- Macros

How do you pass a variable by reference in C++?

- `func(x);`
- `func(*x);`
- ✓ `func(&x);`
- `func(@x);`

What is the purpose of header files like `<iostream.h>`?

- Store arrays
- Declare global variables
- Define main function
- ✓ Provide declarations for library functions

Which symbol marks the start of a preprocessor directive?

- @
- \$
- &
- ✓ #

Which benefit follows encapsulation and data hiding?

- More global variables
- Faster loops
- ✓ Reduced coupling and improved maintainability
- Less code reuse

What does the `const` keyword do in function parameters?

- Enable recursion
- Hide variables

Make variables larger

✓ Prevent modification of arguments

Which of these is not a valid identifier scope?

Function-level scope

Global scope

✓ Hybrid scope

Block-level scope

Lecture 11

How are array elements accessed in C?

Using parentheses

Using dot operator

Using arrow operator

✓ Using index notation like `array[i]`

What is the index of the first element in a C array?

1

-1

Depends on compiler

✓ 0

Which declaration creates an integer array of size 10?

`array int age[10];`

`int[10] age;`

`age int[10];`

✓ `int age[10];`

How do you initialize all elements of an integer array to zero at declaration?

`int a[10] = 0;`

✓ `int a[10] = {0};`

`int a[10] = { };`

`int a = {10};`

What is the result of `int a[5] = {1,2};`?

Error

All values 2

All values 0

✓ `a[0]=1, a[1]=2, rest 0`

Which loop reads N values into an array a of size N?

`for(i=1;i<=N;i++) cin>>a[i];`

✓ `for(i=0;i<N;i++) cin>>a[i];`

`while(i<=N) cin>>a[i];`

`do cin>>a[i]; while(i<N);`

How do you copy one array a to another array b of the same size?

By assignment `b = a;`

Using `memcpy` only

✓ Using a loop: `for (i=0; i<n; i++) b[i]=a[i];`

Using `std::copy` only

What is linear search?

Sorting an array

Binary search

Merging arrays

✓ Scanning each element until a match is found

What is the key difference between linear and binary search?

Linear requires sorted array; binary does not

Uses pointers; whereas binary uses indexes

✓ Linear scans all elements; binary divides the search in half

Binary is always faster for unsorted data

What is the role of the `const` keyword when applied to an array parameter?

Make array dynamic

Allocate memory

To sort array

✓ Prevent modification of array elements

What additional information must you pass when you pass an array to a function?

Data type

Return type

Nothing else

✓ The array's size

Which of the following is not a valid array name?

`int arr_1[5];`

`int a2[5];`

`int myArr[10];`

✓ `int 2arr[5];`

What is the maximum valid index for `int a[10];`?

10

0

-1

✓ 9

Which statement declares a compile-time constant for array size?

`int a[size];`

`size = 10; int a[size];`

`int const a[10];`

✓ `const int size = 10; int a[size];`

What is a C string?

- ☒ A character array terminated by `\0`
- An integer array
- A number
- A function

How do you declare a character array of length 50?

```
char name(50);  
string name[50];  
☒ char name[50];  
char[50] name;
```

What character marks the end of a C string?

- `\n`
- Space
- `/`
- ☒ `\0`

How can you initialize a string at declaration?

```
char s[] = "Hello"; ☒  
char s[5] = {H,e,l,l,o};  
char s = "Hello";  
string s = 'Hello';
```

What happens if a string literal is longer than the array size?

- Supplied with zeros
- ☒ Buffer overflow (undefined behavior)
- Compile error
- Safe truncation

How do you compare two arrays element by element?

- `a == b;`
- Using `strcmp` only
- ☒ With a loop checking `a[i] != b[i]`
- Using `=` operator

What does `for(i=0;i<length;i++) cout << s[i];` do?

- Deletes characters
- ☒ Prints each character one by one
- Initializes the string
- Copies the string

Which escape sequence outputs a newline in `cout`?

- `\t`
- `\\`
- `\"`
- ☒ `\n`

Which escape sequence outputs a tab space?

- `\n`
- `\\`

\"
✓ \t

Which simple sorting algorithm is often taught first?

- Quick sort
- Merge sort
- Radix sort
- ✓ Bubble sort

What search algorithm can be used after sorting an array?

- Linear search only
- Interpolation search
- Random search
- ✓ Binary search

Which header provides the `rand()` function in C++?

- `<string>`
- `<vector>`
- `<iomanip>`
- ✓ `<cstdlib>`

How do you pass an array to a function in C?

- `func(&array);`
- `func();`
- ✓ `func(array, size);`
- `func[];`

What does `rand() % 6 + 1` produce?

- Numbers from -5 to 0
- 0 only
- ✓ A random integer between 1 and 6 inclusive
- A floating-point value

Why might you compare arrays before and after manipulation?

- To slow down code
- To allocate memory
- ✓ To verify correctness of operations
- To hide data

Lecture 13

What is a two-dimensional array?

- A pointer to a pointer
- A string
- A function
- ✓ An array of arrays

How do you declare a 2×3 integer array named `m`?

`int m(2,3);`

```
int m[3][2];  
☒ int m[2][3];  
array<int,2> m[3];
```

Which notation accesses the element in row r , column c ?

`m.c[r]`
`m[r, c]`
`m(r)(c)`
☒ `m[r][c]`

What loop structure prints every element of a 2D array?

Single for loop
☒ Nested for loops
while only
do-while only

How can you initialize a 2×2 matrix at declaration?

```
int m[2][2] = 1,2,3,4;  
☒ int m[2][2] = {{1,2},{3,4}};  
int m[2][2] = {1,2,3,4};  
int m = {{1,2},{3,4}};
```

What escape sequence `\t` is commonly used for when printing matrices?

Newline
Double quote
Backslash
☒ Tab spacing for alignment

Why print the original matrix before the flipped matrix?

To confuse users
To reduce memory usage
To avoid loops
☒ To verify that the manipulation produced the expected result

When passing a 2D array to a function, what additional information must you pass?

No information needed
Row and column indices
☒ At least the number of columns
Data type only

What is the valid range of indices for both dimensions in C arrays?

1 to size
-1 to size-1
1 to size-1
☒ 0 to size-1

How many elements does `int a[3][5];` contain?

8
50

3

✓ 15

Which concept organizes solving a real-world problem into subtasks and functions?

Spaghetti coding

✓ Top-down design (design recipe)

Bottom-up design

Inline coding

What benefit do 2D arrays provide in programming?

Less readability

Faster compilation

Fewer loops

✓ Modeling tables, grids, and matrices

Which of these is not a proper way to declare a 2D array?

`int arr[4][4];`

`int arr[2][3];`

✓ `int arr[][3];` (missing row size in declaration)

`int arr[5][5];`

What does the `\n` escape sequence do when printing a matrix?

Prints a space

Prints a tab

Prints a backslash

✓ Moves output to the next line

Which loop variable names are conventional for row and column in nested loops?

`i` and `i`

`row` and `row`

✓ `i` and `j`

`x` and `x`

Lecture 14

What is a pointer in C?

✓ A variable that stores the address of another variable

A variable that stores a value directly

A function returning a memory address

An array of integers

Which operator is used to obtain the address of a variable?

✓ `&`

`*`

`%`

`@`

How do you declare a pointer to an integer named p?

☒ `int *p;`
`int p*;`
`pointer<int> p;`
`int& p;`

What does the * operator do when applied to a pointer?

Multiplies two values

☒ Dereferences the pointer (accesses the pointed-to value)

Computes the size of the pointer

Declares a new pointer

What is a null pointer?

An uninitialized pointer

A pointer to a random address

☒ A pointer with the value `NULL` (zero)

A pointer that points to itself

Which of these is a correct pointer initialization?

`int *p = x;`
`int p = &x;`
☒ `int *p = &x;`
`int &p = x;`

What is the typical size of a pointer on a 32-bit system?

2 bytes

☒ 4 bytes

8 bytes

Depends on the data type

What happens if you dereference a null pointer?

It returns zero

☒ It causes a segmentation fault (undefined behavior)

It returns the value `NULL`

It prints the memory address

Which declaration defines a pointer to pointer to int?

`int p**;`
`pointer<int*> pp;`
☒ `int **pp;`
`int &*pp;`

What is the effect of the expression `p++` when `p` is an `int*`?

Moves the pointer forward by 1 byte

☒ Moves the pointer forward by `sizeof(int)` bytes

Moves the pointer backward by 1 byte

Leaves the pointer unchanged

Which pointer arithmetic operation is invalid?

Adding an integer to a pointer

Subtracting one pointer from another

Comparing two pointers within the same array

✓ Adding two pointers together

How can you safely indicate an uninitialized pointer?

By setting it to -1

Leaving it uninitialized

✓ By setting it to `NULL`

By pointing it to itself

Which of the following correctly declares a pointer that cannot be changed after initialization?

`const int *p;`

`int const *p;`

✓ `int * const p = &x;`

`const *int p;`

What does the expression `*p = 10;` do when `p` is a valid `int*`?

Changes the pointer's address to 10

✓ Stores the value 10 at the memory location pointed to by `p`

Creates a new integer variable

Causes a syntax error

Which of these best describes pointer arithmetic?

Pointer arithmetic always operates in bytes

Only addition is allowed

✓ Addition or subtraction moves by the size of the pointed-to type

Multiplication and division are supported

Lecture 15

What does the name of an array represent in C?

A variable that can be reassigned

The size of the array

✓ A constant pointer to its first element

An integer value

Which expression gives the address of the third element in `arr`?

`&arr[2]`

✓ Both `&arr[2]` and `arr + 2`

`&arr + 2`

`arr[2]`

What is the relationship between `arr[i]` and `*(arr + i)`?

They differ by one element

One reads from memory, the other writes

An error to use the latter

✓ They are equivalent

Which operation is valid with two pointers to the same array?

Adding them together

Multiplying them

✓ Subtracting them to find the distance

Dividing one by the other

What happens if you try to do `arr++` where `arr` is an array name?

✓ Compilation error (array names are non-modifiable lvalues)

It moves to the next element

It resets the array

It converts to a pointer

Which of these shows pointer-to-array initialization?

`int *p = arr[5];`

`int p = &arr;`

✓ `int *p = arr;`

`int p = arr;`

How do you access the 5th element via pointer `p`?

`p[6]`

`*p[5]`

✓ `p[4]` or `*(p + 4)`

`&p[5]`

What is pointer comparison valid for?

Pointers to different arrays

Pointers to numeric values

✓ Pointers within the same array

Pointers to functions only

Which statement correctly prints all elements of `int a[10]` using a pointer?

`for(p = a; p < a + 10; ++p) cout << a[p];`

✓ `for(p = a; p < a + 10; ++p) cout << *p;`

`for(p = a; p <= a + 10; ++p) cout << *p;`

`for(p = &a; p < &a + 10; ++p) cout << *p;`

How can you pass an array to a function?

By value only

✓ By passing its name and size: `func(arr, 10);`

By assigning it to a pointer inside

By using global arrays only

Which of these is a correct prototype for a function taking an integer array?

`void f(int arr[]);` ✓

`void f(int[]);`

`void f(array<int> arr);`

`void f(int &arr);`

What is the result of comparing two pointers from different arrays?

Always true

Always false

✓ Undefined behavior

Always equal

Which of the following increments the pointer `p` past the end of `arr[5]`?

`p += 4;`

`p = arr + 5;` ✓

`p == arr + 5;`

`p = &arr[5];`

Why is pointer-and-array synergy useful?

To reduce memory usage

To speed up compilation

✓ To allow efficient traversal and manipulation of arrays

To avoid using loops

Lecture 16

Which of these continues the discussion of pointers?

Functions only

Arrays only

Strings only

✓ Multidimensional arrays and advanced pointer usage

How do you declare a 2D array with 3 rows and 4 columns?

`int a[4][3];`

`int a[3,4];`

✓ `int a[3][4];`

`int a[3]*[4];`

What is the valid index range for `a[3][4]`?

1 to 3 and 1 to 4

0 to 4 and 0 to 5

✓ 0–2 for rows, 0–3 for columns

–3 to 3 and –4 to 4

Which nested loop structure prints every element of a 2D array `a`?

`for(i=0;i<3;i++) for(j=0;j<3;j++) cout<<a[i];`

`for(i<3;i++) for(j<4;j++) cout<<a[j][i];`

✓

`for(i=0;i<3;i++)`

`for(j=0;j<4;j++)`

`cout << a[i][j];`

`for(i=1;i<=3;i++) for(j=1;j<=4;j++) cout<<a[i][j];`

How do you access the element in row 2, column 3?

`a[3][2]`

a(2,3)
☒ a[2][3]
*a[2][3]

What is the total number of elements in `int a[5][6];`?

5
6
☒ 30
11

Which statement about pointers to multidimensional arrays is true?

They behave like 1D arrays only
They cannot be passed to functions
☒ Array-of-array semantics still allow pointer arithmetic on the first dimension
They require special pointer types only

How do you pass a 2D array to a function?

By value
☒ `func(int a[][4], int rows);`
`func(&a);`
`func(a, rows, cols);`

Which of these is a valid way to initialize a 2×2 array?

`int m[2][2] = {1,2,3};`
☒ `int m[2][2] = {{1,2},{3,4}};`
`int m[2][2] = 1,2,3,4;`
`int m[][] = {{1,2},{3,4}};`

What symbol represents the tab character when printing a matrix?

`\n`
`\\`
☒ `\t`
`\/`

Which escape sequence moves the cursor to a new line?

`\t`
`\\`
`\"`
☒ `\n`

Why is it helpful to print the original 2D array before manipulating it?

To increase runtime
☒ To verify that your transformation is correct
To hide data
To reduce code readability

Which technique maps well to 2D arrays in real-world problems?

Linked lists
Stacks
☒ Grids, tables, and matrices
Floating-point arithmetic

Which of these is not a valid way to define a jagged array in C?

Using pointers and malloc

Using VLA (variable-length arrays) in C99

✓ Declaring `int a[][];` without sizes

Using a pointer to pointer

Lecture 17

Which header provides C-style string functions like `strcpy`?

`<string>`

`<iostream>`

✓ `<cstring>`

`<cstdlib>`

What does `strlen(s)` return?

Number of bytes allocated

Length including `\0`

✓ Number of characters before the terminating `\0`

Always zero

Which function copies one string into another?

`strcat`

`strcmp`

✓ `strcpy`

`strlen`

What does `strcmp(s1, s2)` return when `s1` equals `s2`?

Positive number

Negative number

✓ Zero

Length of `s1`

Which function concatenates two strings?

`strlen`

`strncmp`

✓ `strcat`

`strcpy`

What does `strstr(haystack, needle)` return?

Length of `needle`

First character of `needle`

✓ Pointer to first occurrence of `needle` in `haystack` or `NULL`

The index of `needle`

Which function converts a string to an integer?

`itoa`

`strtol`

✓ `atoi`

`atof`

Which function converts a string to a floating-point number?

atoi
sprintf
☒ atof
strtod

What does `strncmp(s1, s2, n)` do?

Always compares full strings
Encrypts a string
☒ Compares up to `n` characters of two strings
Concatenates up to `n` characters

Which of these is a safe way to copy strings in C?

`strcpy(dest, src);`
☒ `strncpy(dest, src, n);`
`strcat(dest, src);`
`strcmp(dest, src);`

What is returned by `strchr(s, c)`?

Index of `c`
Boolean
☒ Pointer to the first occurrence of character `c` or NULL
Length of `s`

What happens if you call `strcat` on overlapping buffers?

☒ Undefined behavior
Safe copying
Automatic reallocation
Compiler error

Which function returns a newly allocated string in POSIX?

`strdup`
`strcat`
☒ `strdup`
`strcpy`

Which of these is not a C string function?

`strcpy`
`strlen`
`sprintf`
☒ `vector`

Why are C string functions considered risky?

They are slow
They ignore null terminators
☒ They do not check buffer boundaries (risk of overflow)
They always return zero instead of pointers

Which header file is required for file I/O in C++?

<cstdio>
<string>
☒ <fstream>
<iomanip>

Which class is used to read from a text file?

ofstream
fstream
☒ ifstream
filein

Which class is used to write to a text file?

ifstream
fstream
☒ ofstream
fileout

How do you open a file named "data.txt" for reading?

fstream fin("data.txt", ios::out);
☒ ifstream fin("data.txt");
ifstream fin.open("data.txt", ios::out);
ofstream fin("data.txt");

Which open mode allows reading and writing both?

ios::in
ios::out
☒ ios::in | ios::out
ios::binary

How do you check if a file stream was opened successfully?

Always true
☒ if (!fin) or if (fin.fail())
if (fin == NULL)
if (fin.is_open())

Which member function closes a file?

end()
stop()
☒ close()
finish()

What does the eof() member function indicate?

Start of file
☒ End of file reached
File not found
Buffer full

Which function reads a single line from a file into a string s?

fin >> s;
☒ getline(fin, s);
fin.getline(s);
fin.read(s);

How do you write the integer `x` and a newline to a file `fout`?

```
fout << x; fout.newline();  
fout.write(x); fout.write('\n');
```



cpp

CopyEdit

```
fout << x << '\n';
```

Which mode opens a file for appending at the end?

`ios::in`

`ios::out`

☒ `ios::app`

`ios::trunc`

What happens if you open an existing file with `ios::trunc`?

☒ Its contents are discarded (file is truncated)

It appends instead

It fails to open

It reads only

Which flag specifies binary mode?

`ios::text`

☒ `ios::binary`

`ios::raw`

`ios::none`

How can you write formatted output (like `printf`) to an `ofstream`?

```
fprintf(fout, ...)
```

☒ Use `ostringstream` then `fout << oss.str();` or use `sprintf` into a buffer then write

```
printf(fout, ...)
```

```
fout.printf()
```

What is the basic workflow for text file handling in C++?

Declare variables → close file → open file → read/write

Open file → read/write → close file → check errors

☒ Declare stream → open file → read/write data → close file → handle errors

Open file → declare stream → read/write → forget to close

Lecture 19

Which file access mode lets you move the read/write pointer to any position?

Sequential access only

☒ Random access using streams opened with `ios::in` | `ios::out` | `ios::binary`

Buffered access

Text-only access

What does the `tellg()` function return?

Current put pointer position

Number of bytes read

✓ Current get pointer position in the input stream

Total file size

Which function sets the get pointer to a specified location?

`seekp()`

`tellp()`

✓ `seekg(position)`

`goto(position)`

What is the counterpart of `seekg()` for the output pointer?

`seekg()`

`tellg()`

✓ `seekp()`

`tellp()`

Which `fstream` object supports both reading and writing?

`ifstream`

`ofstream`

✓ `fstream`

`filebuf`

How can you efficiently read or write large blocks of data?

One byte at a time with `get()/put()`

✓ Using `read(buffer, size)` and `write(buffer, size)`

Using `>>` and `<<` operators only

Using `getline`

What happens when you copy a file in reverse order?

Nothing; it fails

✓ You seek from end toward beginning and write blocks backward

You use a special reverse mode

You can't reverse files

Why insert data in the middle of a file by random access instead of rewriting the entire file?

It's slower

✓ It avoids reading and rewriting the entire file, improving performance

It always corrupts the file

It's the only supported method

What must you check after opening a file for random access?

That it's text mode

✓ That the stream is valid (`if (!fs) ...`)

That `seekg` succeeded

That it's on USB drive

Which function reports the current put pointer position?

`tellg()`

✓ `tellp()`
`seekg()`
`seekp()`

What is the effect of opening a file with `ios::app` when also using random access?

It truncates the file

✓ All output is appended, but pointer moves to end before each write

It allows only input

It fails to open

When inserting data mid-file, why might you use a temporary file?

To avoid using `seekg()`

✓ To simplify logic: copy up to insert point, write new data, then copy remainder

To make it sequential only

To convert to binary

Which flag combination opens a binary file for both input and output?

`ios::binary` only

`ios::in | ios::binary`

`ios::out | ios::binary`

✓ `ios::in | ios::out | ios::binary`

What is the primary advantage of random access files?

They compress data

They are human-readable

✓ You can jump directly to needed records without full scan

They require no streams

How do you copy a file's contents to another file in reverse?

By reading forward and writing backward

By using `std::reverse_copy`

✓ By seeking from end toward start in blocks and writing to the output

By using `copy_r` utility

Lecture 20

Which keyword is used to define a structure in C++?

`union`

`class`

`object`

✓ `struct`

What must end a `struct` declaration?

A comma (,)

A closing brace without punctuation

✓ A semicolon (;) after the closing }

An extra }

How do you access member age of struct Person p?

p->age
p.age; ✓
Person.age;
age(p);

Which syntax initializes a struct at declaration?

Person p = age:25;
✓ Person p = { "Alice", 25 };
Person p("Alice",25);
Person p = .Alice.,25;

How do you pass a struct to a function by value?

void f(Person &p);
void f(const Person p);
✓ void f(Person p);
void f(p Person);

What does sizeof(MyStruct) return?

Number of members
Sum of member lengths as text
✓ Total bytes occupied by the struct (including padding)
Number of fields

How do you define an array of 10 Person structs?

Person[10] arr;
struct Person arr;
✓ Person arr[10];
array<Person> arr(10);

Which keyword defines a union?

struct
class
✓ union
combine

What is a key difference between struct and union?

Structs share memory; unions do not
✓ Unions share the same memory for all members; structs allocate separate memory
Structs can't have functions; unions can
Unions require template

Why use a union?

For multiple inheritance
To store only one of several types in the same memory location ✓
To speed up compilation
To avoid padding

How do you access a struct member via a pointer p?

p.age
(*p).age()
✓ p->age
->p.age

Which of the following is invalid inside a `struct`?

- Member variables
- Member functions (in C++)
- Public/private specifiers
- ☒ Templates (pre-C++11 for POD structs)

What happens if you omit the struct's trailing `;`?

- ☒ Compilation error
- It adds one automatically
- Runtime error
- Linker warning

How do you zero-initialize an array of structs?

- `Person arr[5] = 0;`
- `Person arr = {};`
- ☒ `Person arr[5] = {};`
- `Person arr[5] = { 0 };`

What is the purpose of `typedef struct { ... } Name;`?

- Deletes the struct
- Renames a function
- Defines a union
- ☒ Gives the struct a type alias `Name` without needing `struct Name`

Lecture 21

Which operator performs bitwise AND?

- `&&`
- AND
- ☒ `&`
- and

Which operator is bitwise OR?

- `||`
- +
- OR
- ☒ `|`

What does the bitwise XOR operator `^` do?

- Logical OR
- Assignment
- ☒ Sets bits where operands differ
- Sets bits where both are 1

Which operator inverts all bits of its operand?

- !
-
- ☒ `~`
- `^`

How do you left-shift bits of x by 2 positions?

```
x >> 2;  
x <<< 2;  
x = << 2;  
✓ x << 2;
```

Which operation clears bit n of x ?

```
x &= (1 << n);  
x |= (1 << n);  
x ^= (1 << n);  
✓ x &= ~(1 << n);
```

How do you set bit n of x ?

```
x &= ~(1 << n);  
✓ x |= (1 << n);  
x ^= (1 << n);  
x << n;
```

What does $x \wedge= (1 \ll n)$; accomplish?

Set bit n only

✓ Toggle (flip) bit n

Clear bit n permanently

Shift x by n

Which data type is best for bitwise operations?

```
float  
double  
int  
✓ unsigned int
```

How can you test if bit n of x is set?

```
if (x == n)  
if (x & n)  
✓ if (x & (1 << n))  
if (x << n)
```

What is a “mask” in bit manipulation?

A file mode

✓ A bit pattern used to isolate or modify specific bits

A debugger flag

An encryption key

Which operator shifts bits right, filling with zeros for unsigned types?

```
>>>  
>>=  
>=  
✓ >>
```

What is a “bit flag”?

A physical switch

A file descriptor

✓ A single bit representing a boolean condition

A network protocol

Why use unsigned integers for bitwise flags?

- | They cannot be shifted
- ✓ To avoid sign-extension issues on right shifts
- They take more memory
- They are slower

Which expression clears the lowest set bit of x ?

- $x \& (x + 1);$
 - $x \mid (x - 1);$
 - ✓ $x \& (x - 1);$
 - $x \wedge (x - 1);$
-

Lecture 22

Which operator both shifts left and assigns in one step?

- $\ll$
- $\ll=$ ✓
- $\leq$
- $=$

What keyword starts a nested decision in C++?

- else
- elif
- ✓ if
- switch

Which loop guarantees at least one execution of its body?

- for
- while
- ✓ do { ... } while(condition);
- if

Which keyword exits a switch case and continues execution after the switch?

- exit
- continue
- ✓ break
- fallthrough

What does `continue` do inside a loop?

- Stops the program
- Ends all loops
- ✓ Skips the rest of the current iteration and proceeds with the next
- Jumps to a label

Which is the correct syntax for a switch statement on x ?

- switch x { ... }
- switch(x) { ... } ✓
- switch(x): ... end switch
- iff(x) { ... }

What is the first step in the design recipe for solving problems?

Code immediately

✓ Analyze and understand the problem

Write test cases

Declare variables

Which assignment operator multiplies `x` by 5 and stores the result in `x`?

`x = x * 5;`

✓ `x *= 5;`

`x = 5x;`

`x + =5;`

What type is best for storing text characters?

`int`

`float`

✓ `char` (or `char[]`)

`double`

Which declaration makes a variable read-only after initialization?

`int x;`

`static int x;`

✓ `const int x;`

`volatile int x;`

What loop header runs from `i = 0` while `i < n`, incrementing `i` each time?

`while(i<n)`

`do i++ while(i<n)`

`for(n;i;)`

✓ `for(int i = 0; i < n; i++)`

Which structure is used for multi-way branching on an integer?

`if ... else`

`for`

✓ `switch ... case`

`while`

How do you write a two-way decision?

`if (cond) stmt` ✓

`switch(cond): stmt`

`loop(cond) stmt`

`do cond then stmt`

What is the purpose of the `break` statement inside loops?

Ends the program

Skips to next iteration

✓ Exits the loop immediately

Restarts the loop

Which control construct should you use for testing multiple discrete values of a variable?

`while`

`for`

✓ switch
goto

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